

GEOGRAPHY

PHYSICAL GEOGRAPHY



CLIMATOLOGY

MODULE – 35

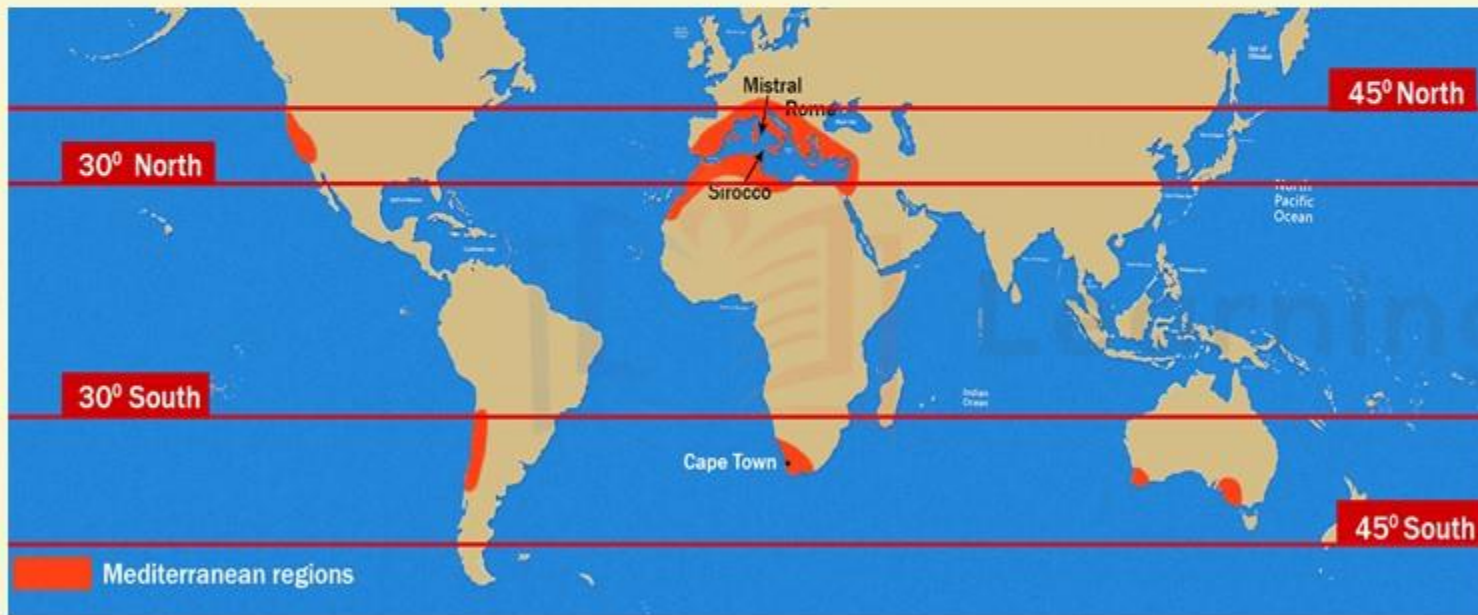
WORLD CLIMATIC TYPES
THE MEDITERRANEAN CLIMATE
(PART- 1)

WORLD CLIMATIC TYPES

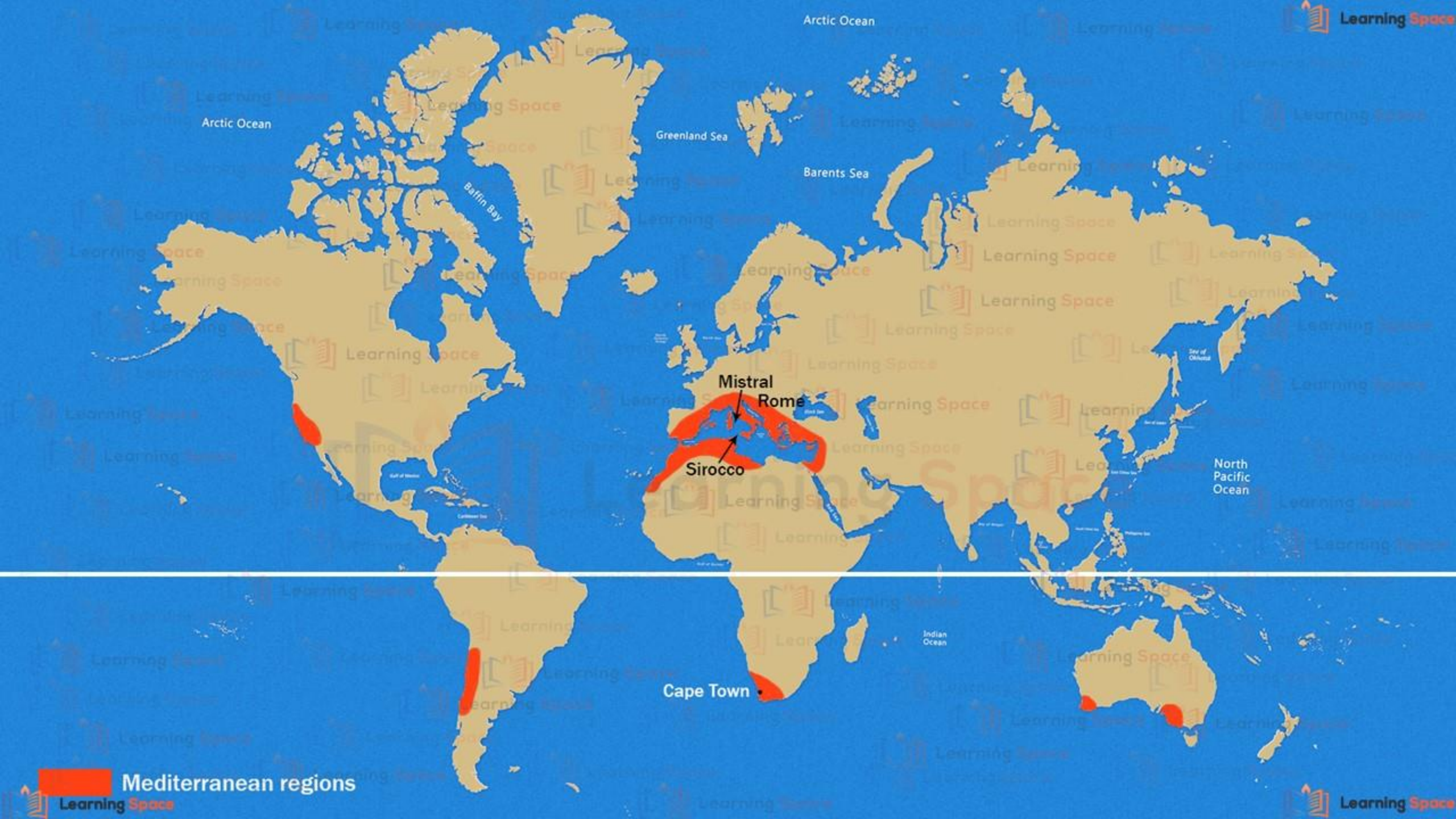
4

THE WARM TEMPERATE WESTERN MARGIN (MEDITERRANEAN) CLIMATE

DISTRIBUTION



- 1 The warm temperate Western Margin Climate is found in relatively few areas in the world.
- 2 They are entirely confined to the western portion of continental masses, between 30° and 45° north and south of the equator.
- 3 The basic cause of this type of climate is the shifting of wind belts.



Mediterranean regions

4

THE WARM TEMPERATE WESTERN MARGIN (MEDITERRANEAN) CLIMATE

DISTRIBUTION



4

Though the area around the Mediterranean Sea has the greatest extent of this type of 'winter rain climate', and gives rise to the more popular name Mediterranean Climate, the best developed form of this peculiar climatic type is in fact, found in Central Chile.

WORLD CLIMATIC TYPES

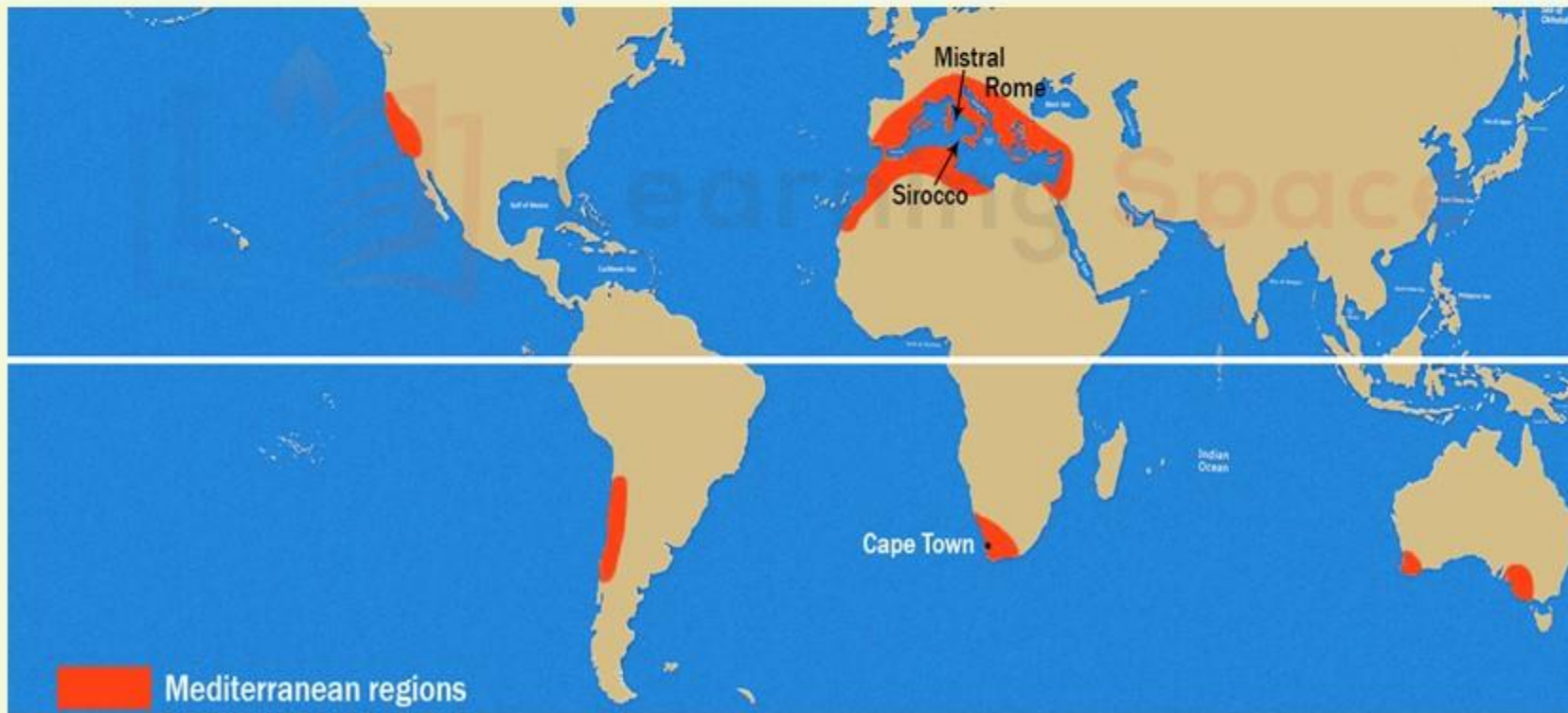
4

THE WARM TEMPERATE WESTERN MARGIN (MEDITERRANEAN) CLIMATE

DISTRIBUTION

5

Other Mediterranean regions include California (around San Francisco), the south-western tip of Africa (around Cape Town), southern Australia (in southern Victoria and around Adelaide, bordering the St. Vincent and Spencer Gulfs), and south-west Australia (Swanland).



4

THE WARM TEMPERATE WESTERN MARGIN (MEDITERRANEAN) CLIMATE

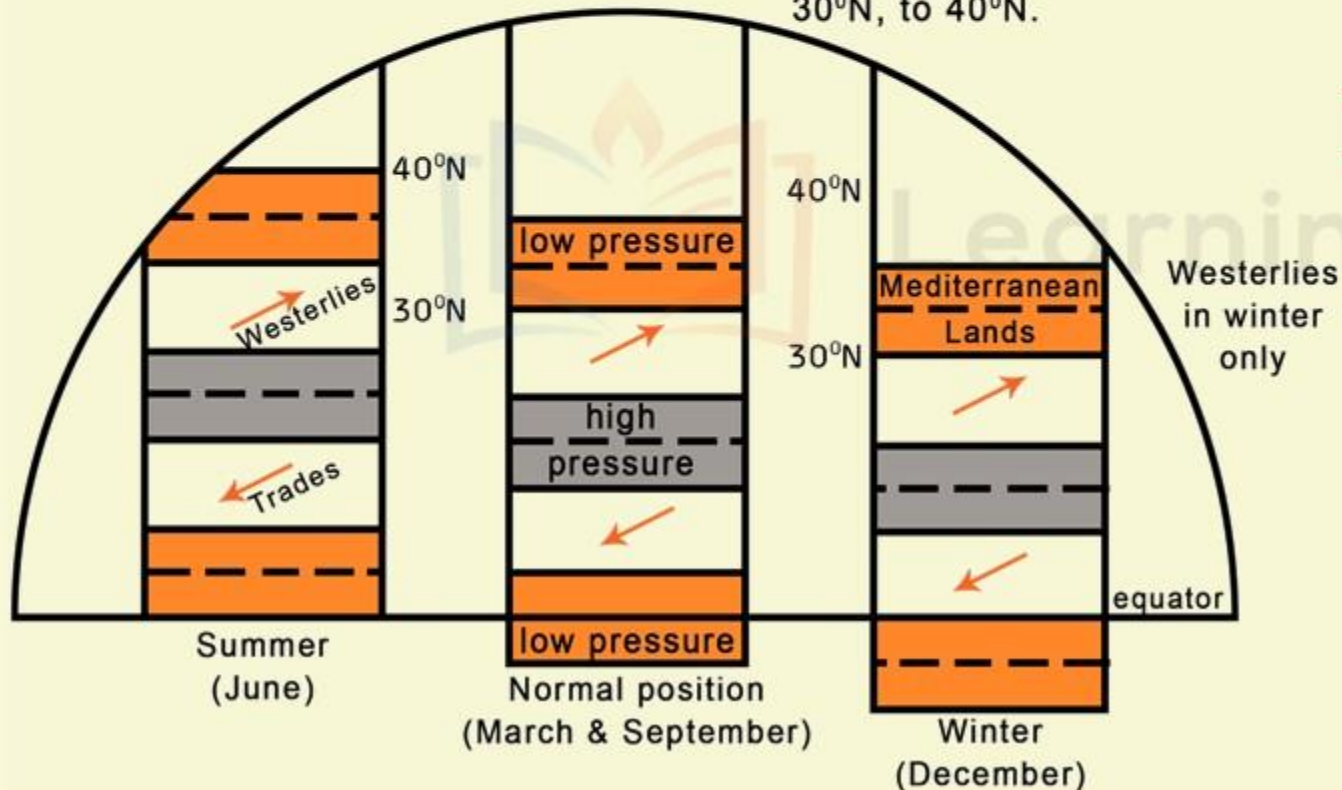
CLIMATE

The Mediterranean type of climate is characterized by very distinctive climatic features.

1

A DRY, WARM SUMMER WITH OFF-SHORE TRADES

Shifting of wind belt in temperate latitude
30°N, to 40°N.



1

In summer when the sun is overhead at the tropic of Cancer, the belt of influence of the Westerlies is shifted a little polewards. Rain bearing winds are therefore not likely to reach the Mediterranean lands.

4

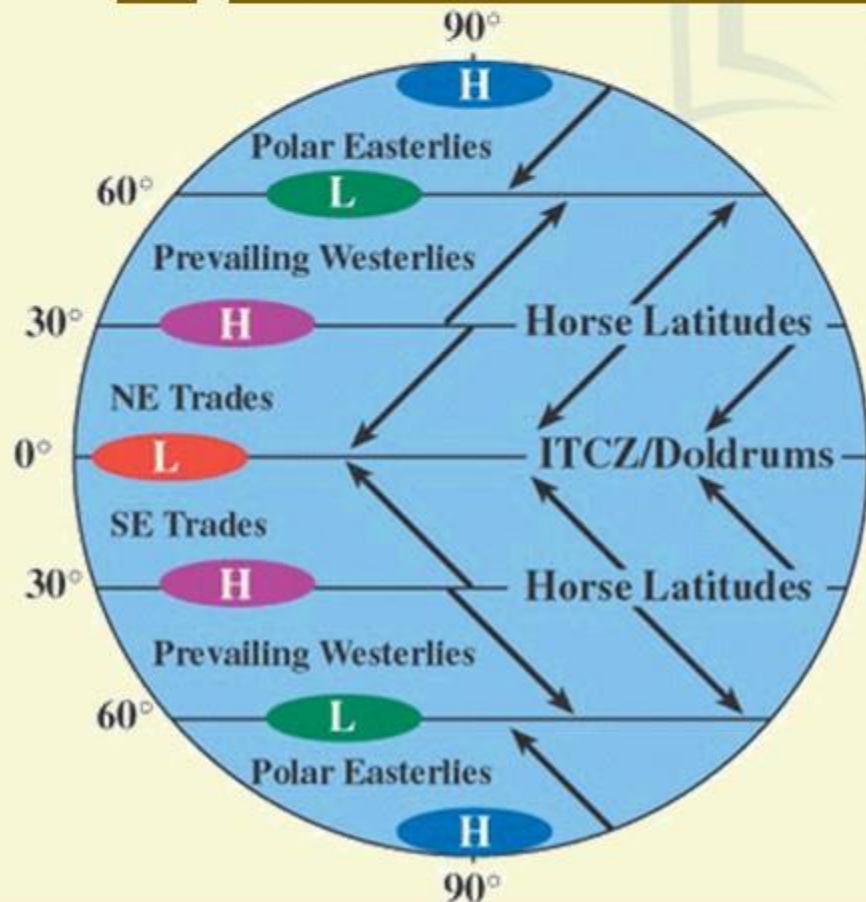
THE WARM TEMPERATE WESTERN MARGIN (MEDITERRANEAN) CLIMATE

CLIMATE

The Mediterranean type of climate is characterized by very distinctive climatic features.

1

A DRY, WARM SUMMER WITH OFF-SHORE TRADES



2

The prevailing Trade Winds are off-shore and there is practically no rain.

3

The air is dry, the heat is intense and the relative humidity is low. Days are excessively warm and in the interiors, prolonged droughts are common.

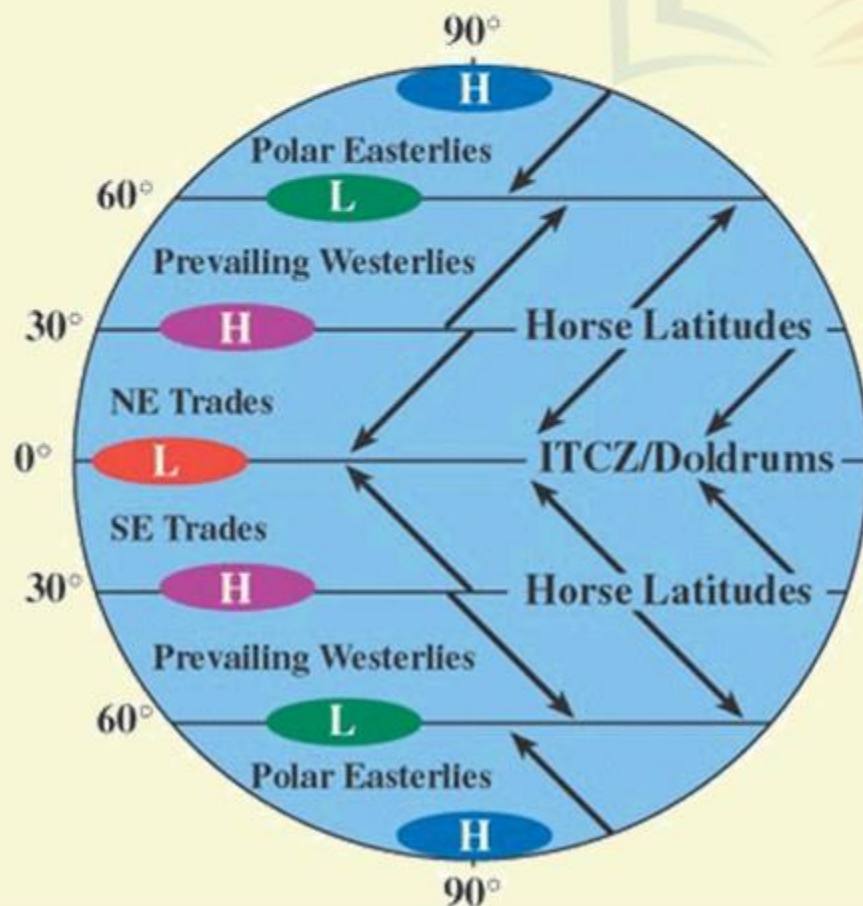
4

THE WARM TEMPERATE WESTERN MARGIN (MEDITERRANEAN) CLIMATE

CLIMATE

2

A CONCENTRATION OF RAINFALL IN WINTER WITH ON-SHORE WESTERLIES



1

The Mediterranean lands receive most of their precipitation in winter when the westerlies shift equator-wards.

2

In the northern hemisphere, the prevailing on-shore westerlies bring much cyclonic rain from the Atlantic to the countries bordering the Mediterranean Sea.

3

This is the rainy season and is the most outstanding feature of the Mediterranean Climate.

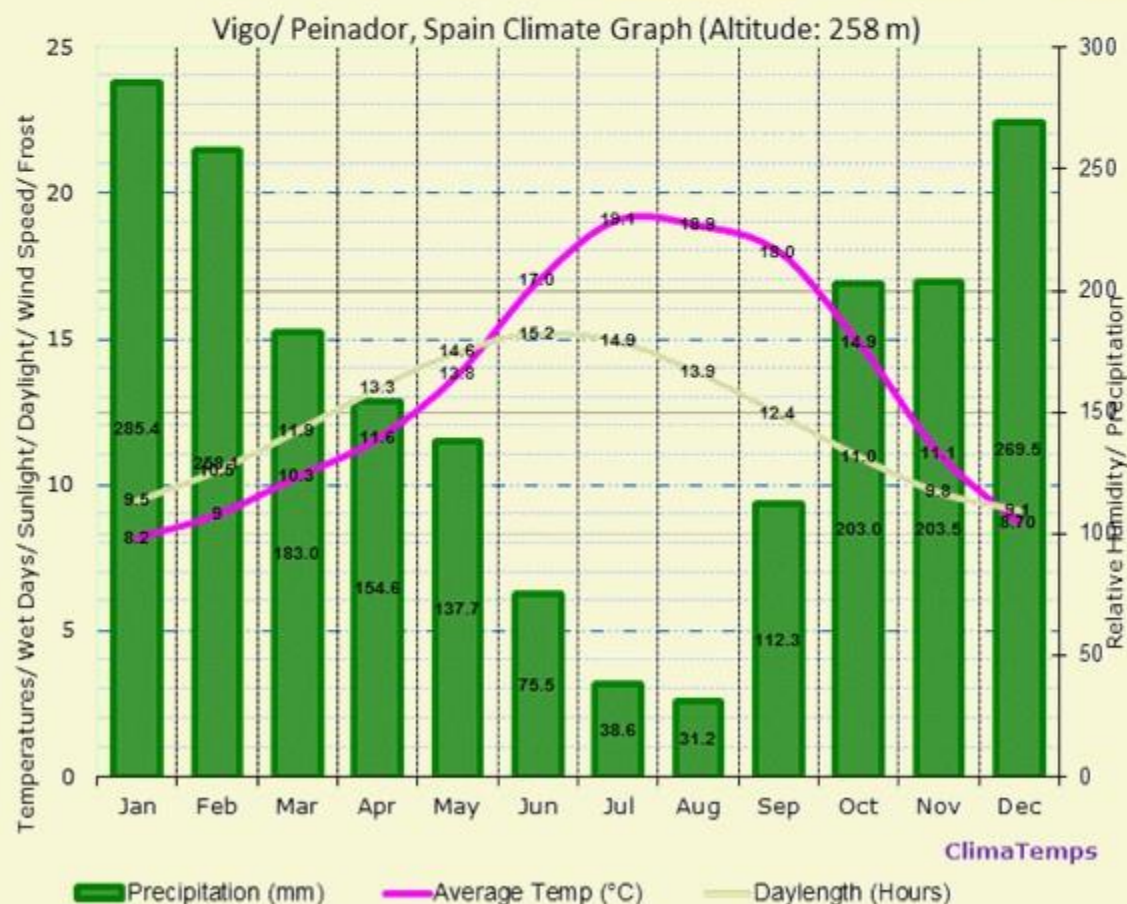
4

THE WARM TEMPERATE WESTERN MARGIN (MEDITERRANEAN) CLIMATE

CLIMATE

2

A CONCENTRATION OF RAINFALL IN WINTER WITH ON-SHORE WESTERLIES



4

In almost all other climatic types maximum rain comes in summer. The mean annual rainfall is normally taken as 25 inches.

5

The rain comes in heavy showers and only a few days with bright sunny period is between them. This is another characteristic feature of the Mediterranean winter rain.

6

Snow rarely occurs on lowlands and coastal districts and even if it does fall on the highlands, it is moderate and is a source of water supply for hydro-electric power generation and for irrigation.

4

MEDITERRANEAN CLIMATE

CLIMATE

3

BRIGHT, SUNNY WEATHER WITH HOT DRY SUMMERS AND WET, MILD WINTERS



1

Considering its mid-latitude position, the Mediterranean regions have a very favorable climate, unrivalled by any other climatic regions.

2

The climatic features are transitional between those of the Trade Wind Hot Desert in the South and the Cool Temperate Maritime Climate in the North.

3

Summers are warm and bright and winters are so mild and cool that many tourists come at all times of the year. The sky is almost cloudless and sunshine is always abundant.

4

MEDITERRANEAN CLIMATE

CLIMATE

4

THE PROMINENCE OF LOCAL WINDS AROUND THE MEDITERRANEAN SEA

1

Many local winds, some hot, others cold are common around the Mediterranean Sea.

2

The causes are many and varied. The topography of the region with the high Alps in the north, the Sahara Desert in the south, continental interiors in the east and the open Atlantic on the west give rise to great differences in temperature, pressure and precipitation.

3

These winds varying in strength, direction and duration affect the lives, crops and activities of the people there.



WATCH THIS
MODULE AGAIN

GEOGRAPHY

PHYSICAL GEOGRAPHY



CLIMATOLOGY

MODULE – 20

CLASSIFICATION OF WINDS

- LOCAL WINDS(CONTD..)



PROBABLE QUESTION FOR MAINS

Q 3 WHAT ARE LOCAL WINDS? HOW DO THEY AFFECT THE LOCAL WEATHER CONDITIONS AND HUMAN HABITATS? **(10 Marks) (150 words)**

HINTS:

1. Straight forward question. Go with the theory part we have dealt.
2. Define local winds.
3. Thermal factors and topographical factors.
4. Write second half answer with the NAMES AND PLACES of local winds.
5. Chinook, Foehn- Snow eater; 4 degree Celsius drop; wheat sowing; grapes ripening.
6. Sirocco- Blood Rain; Injurious to agri crops.
7. Blizzard- Cold snow storm even upto southern states of US.
8. Similarly, Harmattan also can be quoted as an example.

4

MEDITERRANEAN CLIMATE

CLIMATE

4

THE PROMINENCE OF LOCAL WINDS AROUND THE MEDITERRANEAN SEA

a) SIROCCO

1

This is a hot, dry dusty wind which originates in the Sahara desert.

2

Though it may occur at any time of the year, it is most frequent in spring and normally lasts for only a few days.

3

The sirocco blows in a southerly direction from the desert interiors into the cooler Mediterranean sea.

4

After crossing the Mediterranean sea, the sirocco absorbs moisture from it.



4

MEDITERRANEAN CLIMATE

CLIMATE

4

THE PROMINENCE OF LOCAL WINDS AROUND THE MEDITERRANEAN SEA

a) SIROCCO

OTHER NAMES OF SIROCCO

NAME OF THE WINDS	PLACES
Khamsin	Egypt
Chilli	Tunisia
Gibli	Lybia

5

Even then, it is still hot and dry with a temperature of over 105° F. Its scorching breath withers vegetation and crops.

6

The damage is particularly serious when it comes at the times during which vines and olives are in blossom.

7

The sirocco is so prominent that it is called by many other local names, such as Chili in Tunisia, Ghibli in Libya, Leveche in Spain, Khamsin in Egypt and Malta.

4

MEDITERRANEAN CLIMATE

CLIMATE

4

THE PROMINENCE OF LOCAL WINDS AROUND THE MEDITERRANEAN SEA



b) MISTRAL

1

In contrast, the Mistral is a cold wind from the north, rushing down the Rhone valley in violent gusts between 40 and 80 miles per hour.

2

The velocity of the Mistral is intensified by the funnelling effect in the valley between Alps and the Central Massif, and in the extreme cases trains may be derailed and trees uprooted.

Mistrals are a cold, northerly wind flowing into the Gulf of Lion from the southern coast of France.

4

MEDITERRANEAN CLIMATE

CLIMATE

4

THE PROMINENCE OF LOCAL WINDS AROUND THE MEDITERRANEAN SEA



b) MISTRAL

3

In winter when the Mistral is most frequent the temperature of the wind may be below freezing-point, though the sky may be clear and cloudless.

4

As a protective measure, many of the houses and orchards of the Rhone valley have thick rows of trees and hedges planted to shield them from the Mistral.

4

MEDITERRANEAN CLIMATE

CLIMATE

4

THE PROMINENCE OF LOCAL WINDS AROUND THE MEDITERRANEAN SEA



c) BORA

1

A similar type of cold north-easterly wind experienced along the Adriatic coast is called the Bora.

2

Like the Mistral, it is caused by a difference in pressure between continental Europe and the Mediterranean.

3

MEDITERRANEAN CLIMATE

CLIMATE

4

THE PROMINENCE OF LOCAL WINDS AROUND THE MEDITERRANEAN SEA



BORA WIND (BURJA)
ADRIATIC SEA (KVARNER, CROATIA)

c) BORA

3

This usually occurs in winter, when the atmospheric pressure over continental Europe is higher than that of Mediterranean Sea.

4

The bora thus blows outwards into the Mediterranean. This dry, icy wind is even more violent than the Mistral and speeds of over 100 miles per hour have been recorded. During strong Boras, ships may be blown aground and agricultural lands devastated.

4

MEDITERRANEAN CLIMATE NATURAL VEGETATION

- 1** In a land where half the year is dry, one cannot expect the natural vegetation to be luxuriant.



- 2** Trees with small broad leaves are widely spaced and never very tall.
- 3** The absence of shade is a distinct feature of Mediterranean lands.
- 4** Growth is slow in the cooler and wetter season, even though more rain comes in winter.
- 5** Growth is thus almost restricted to autumn and spring when the temperature is higher and moisture is 'just sufficient'. The long summer drought checks the growth.

4

MEDITERRANEAN CLIMATE

NATURAL VEGETATION

6 One fact is clear, plants whether trees or shrubs, evergreen or deciduous, have to devise ways of adapting themselves to a climatic environment with such a marked rhythmic recurrence of rain and drought.

7 In many ways, the vegetational responses to climatic demands in the Mediterranean are similar to those of the adjacent deserts and the Savanna further south.

8 Plants are in a continuous struggle against heat, dry air, excessive evaporation and prolonged droughts.

9 They are in short xerophytic, a word used to describe the drought-resistant plants in an environment deficient in moisture.

WE EXPRESS OUR SINCERE THANKS FOR VIEWING THIS VIDEO

Presented by



Learning Space

For Suggestions:

suggestions@learningspace.in

To Contact us:

info@learningspace.in

OUR TEAM

G. V. Rao
K. Srikanth
D. Sunil Kumar
L. V. Krishna
Y. Deepthi

Amrita Naidu
M. Binukrishna
G. Ravi Babu
D. Joji
K. Victor Babu

Visit us at:

www.learningspacedigital.com

0 8 6 6 -2 4 4 4 4 7 2
0 9 8 4 9 9 4 2 2 9 9